

STRAY SAGA

Production Blueprint & Game Requirements Specification

DOCUMENT TYPE: GDD / PRD
SPECIFICATION

INDUSTRY TARGET:
PRODUCTION STANDARD

PROJECT STATE: CORE ALPHA
SYSTEM

This document establishes the official product architecture, technical implementation guidelines, and structural requirements for **Stray Saga**. It serves as a single source of truth for engineering and design teams to build scalable visual and tactical mechanics matching professional industry workflows.

1. UI & VISUAL POLISH ARCHITECTURE (UIX)

The visual framework bridges the gap between static asset layers and real-time game loops. It guarantees feedback states stay perfectly synchronized across all menu interfaces, narrative instances, and exploration spaces.

REQ-UIX-01: DYNAMIC MAP TOKEN OVERLAYS & MOOD ANIMATORS

The system must dynamically alter the active exploration sprite avatar's state machine, timeline parameters, and transformation metrics based on underlying companion status properties.

- **Thermal Distress State:** When a companion's **Warmth** property drops below a critical safety floor ($< 20\%$), apply an iterative offset matrix or coordinate jitter vector to simulate high-frequency shivering. Concurrently trigger a local ice-shimmer particle loop overlay.
- **Exhaustion Walk Modification:** When the **Energy** pool falls under 30%, scale back the baseline walking animation frame playback and interpolation configuration by exactly 50% to visually render a heavy, sluggish crawl.
- **Orientation Constraints:** Ensure character orientation maps accurately to active tile paths, preventing flipping anomalies across traversal nodes.

REQ-UIX-02: HEAD-UP DISPLAY (HUD) DASHBOARD INTEGRATION

Implement a persistent, lightweight status tracking container within the active map viewport coordinates to maximize data readability throughout continuous gameplay phases.

- **Interface Layout Anchors:** Lock a high-contrast dashboard panel directly to a top-level viewport coordinate boundary to bypass interactive collision boxes.
- **Icon Configuration Mapping:** Ensure unified graphic symbols correlate instantly with target stats:
 - *Energy:* Visualized via a custom high-contrast lightning bolt/fish-skeleton segmented progress track.
 - *Warmth:* Monitored via a gradient-mapped thermal fire flame icon container.
 - *Trust:* Displayed through a fractional fill-state pixelated heart container tracker.
- **Reactive Pipeline Bounds:** Bind HUD progression visuals directly to character status model events to keep updates immediate.

REQ-UIX-03: UNIFIED SKEUOMORPHIC & EDITORIAL DESIGN SYSTEM

Enforce strict visual layout synchronization rules to align standard game interaction UI tools perfectly with core parchment assets, eliminating conflicting modern styles.

- **Action Button Standardization:** Re-skin generic rectangular button objects (e.g., "Next", "Read Monologue") with custom-wrapped skeuomorphic textures that match primary button containers.
- **Border Accent Dressing:** Apply sharp 1px clean pixel boundaries or gold-leaf filigree vector framing to structural layout backdrops to safely contain oversized illustrated assets.

2. ADVANCED RPG MECHANICS EXPANSION (RPG)

This module changes companion choices into active survival archetypes, adding deep tactical choices, resource constraints, and mechanical trade-offs across every playthrough.

CLASS SYSTEM ATTRIBUTE ALIGNMENT

COMPANION ID	DESIGNATED ARCHETYPE	STARTING METRIC ADVANTAGE	INHERENT HIDDEN FEATURE / TRAIT
Luna	The Adaptable Forager	+20% Maximum Base Energy	Foraging scale multipliers apply when looting trash bin nodes.
Grey	The Stealth Cautious	+15% Starting Warmth Rating	Reduces alert detection parameters of aggressive map entities.
Shadow	The Street-Smart Rogue	+10% Baseline Agility Profile	Unlocks alternative pathway options across blocked obstacle bounds.
Cookie	The Charming Innocent	+20% Base Human Trust Pool	Gains double the flat points from successful human dialogue checks.

REQ-RPG-01: TURN-BASED ACTION POINT TRAVERSAL ENGINE

Convert standard node-to-node pathing choices into a calculated tactical grid puzzle managed by strict character resource consumption rules.

- **Action Point Calculations:** Bind character Action Points (AP) cleanly to the active Energy pool. Moving across standard clear tiles costs 10 AP. Moving through complex terrain blocks (e.g., fence climbing, wet mud puddles) requires a higher AP cost.
- **Weather Phase Integration:** Advancing past a path length sequence ticks the active turn cycle tracker forward. Every turn phase change inflicts an environmental drain that reduces the Warmth rating. Reaching zero warmth triggers a permanent "Hypothermia" penalty state that splits your AP turn recovery rate in half.

REQ-RPG-02: CONTEXTUAL SKILL-CHECK RESOLUTION MATRIX

Replace static event prompt sequences on map interaction positions with active multi-choice skill checks verified against current companion properties.

Sample Script Node: The Aggressive Alley Cat Encounter

- *[FIGHT] Force your way through. (Validation check: Requires > 60% Energy. Outcome: Path clears on success; drops Warmth by 30% on failure.)*
- *[SNEAK] Slip behind the trash bins. (Validation check: Automated success for Shadow archetype. Alternative archetypes trigger an Agility check.)*
- *[BEG] Cry out for human rescue. (Validation check: Requires > 40% human Trust. Outcome: Clears threat tile and spawns a food drop.)*

REQ-RPG-03: MODULAR INVENTORY & FORAGING LOOP

Provide a secure item collection management layout to store, process, and track temporary stat boosts and equippable modifiers.

- **Storage Volume Caps:** Configure a hard 2-slot or 3-slot item array limit within the runtime data schemas.
- **Functional Item Classifications:**
 - *Consumables:* Items that apply instant metric additions on use (e.g., "Discarded Fish Skeleton" injects +25% Energy; "Dry Cardboard Sheet" instantly returns +30% Warmth safety margin).
 - *Accessories:* Non-perishable gear that modifies active status traits while equipped (e.g., "Lost Red Collar" permanently scales up all future human Trust point gains by 10%).

REQ-RPG-04: ROGUE-LITE META-PROGRESSION PIPELINE

Build an enduring progression feedback framework around game-over loops to give players persistent value across successive exploration attempts.

- **Run Defeat Processing:** When character Energy lands at 0% before reaching Eleanor's cottage steps, calculate all accumulated **Trust** scores and convert them into a permanent global currency called **Hope**.
- **Meta Upgrades:** Create a persistent main menu shop interface that reads your **Hope** wallet balance to let players permanently unlock global survival upgrades (e.g., +5% Starting Warmth floor, or revealing food resource node positions from the start of Act I).

3. NARRATIVE & SENSORY DELIVERY (SND)

Sensory delivery guidelines ensure that changes in game sound and pacing respond immediately to environmental stress, increasing immersion during heavy narrative moments.

REQ-SND-01: LAYERED CROSSFADE SOUNDSCAPE ARCHITECTURE

Expand the audio engine into an interactive mixing module that automatically controls track configurations and volume weight groups based on environmental zone coordinates.

- **The Alley Stage Mix:** Run low, wind-howling sub-drones combined with filtered remote city traffic hums.
- **Discovery Event Layer:** Crossfade toward bright, high-register acoustic guitar plucks or synthesized woodwinds to highlight safe areas.
- **The Cottage Threshold Layer:** Shift master outputs smoothly into warm piano keys mixed with close-mic wood hearthfire crackles once the final node conditions are met.

REQ-SND-02: PRECISION TYPEWRITER MONOLOGUE ANIMATION

Replace instantaneous dialogue text injections within narration frames with an adjustable character-by-character typewriter reveal loop.

- **Letter Pacing Controls:** Render text strings at a steady, controlled rate of 25ms to 35ms per individual letter object.
- **Interaction Bypass:** Allow players to bypass active character text delays and instantly fill out the dialogue pane by clicking anywhere on the screen container.